

The MESO Project: A Primer

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This document serves as an introduction to the MESO project. MESO (Modelling Eddies in the Southern Ocean) is an effort to model the entire Southern Ocean at eddy-resolving resolution. No technical or modelling details are discussed here. Rather, we present an overview of the project, with a discussion of its overall scientific importance and role in the Lab.

1 What is MESO?

The ocean circulation is an integral component of the climate system. The ocean transports roughly one third to one half of the total meridional heat flux in the climate system. It may also be responsible for modulating climate variability on the multi-year and longer timescales. Thus, if we are to model climate accurately, we must model the ocean system accurately, and the MESO project is a large-scale effort toward this end. The main features of the modelling effort are as follows:

1. The modelling domain is the entire Southern Ocean, from the equator to close to the Antarctic continent.
2. The model resolution is ‘eddy resolving’, within the limits of computational resources. Currently, simulations are taking place at about $1/4^\circ$ horizontal resolution, with about 40 vertical levels. We intend to increase this resolution as practicable with the new computer. We will also perform a suite of simulations at lower resolution, with eddy parameterization schemes as appropriate.
3. Two models with quite different formulations are being used in this effort: MOM, a z-coordinate model, and HIM, an isopycnal model.
4. The atmosphere is specified. Winds and surface fluxes are taken from NCEP reanalyses, in addition to a relaxation to observed surface values of temperature and salinity. Currently, these fields are time-averaged, with no seasonal cycle, and there is no ice.
5. Once the control integrations have stabilized, we will perform experiments to explore how the transport properties and water mass formation in the Southern Ocean change with forcing parameters, for example a changing wind field as predicted in some global warming simulations at GFDL. The predictability of an eddying ocean may also be investigated. Plainly we cannot integrate the model long enough for the abyss to fully equilibrate. Rather, we estimate that integrations of order 20 years will suffice to give a first order indication of the effects of changing parameters.

2 What is its importance?

The importance of the project arises in three general areas:

1. To understand the Southern Ocean in particular.
2. To understand the role of eddies in the general circulation, and hence help parameterize these effects in the ocean models we use now for coupled simulations.
3. To gain experience in eddying calculations, as these will become increasingly used in short-term climate projections as computer power increases.

We deal with these issues in more detail in the following subsections.

2.1 The Southern Ocean

Why are we focussing on the Southern Ocean?

1. The Southern Ocean dynamics may be responsible for setting the global mean stratification (c.f., Toggweiler and Samuels 1998; Vallis 1999; Gnanadesikan 1999; Gnanadesikan and Hallberg 2000), and the eddy field in the Southern Ocean may be of paramount importance in this. (Fig. 1 shows a typical snapshot of the circulation, showing the effects of resolution on the circulation near the Agulhas.) Models which fail to accurately represent the sensitivity of Southern Ocean eddy fluxes to the mean state of the ocean and vice versa may err in predicting the response of the ocean to changing atmospheric forcing.
2. Similarly, it seems that the Southern Ocean may also play a critical role in the global carbon cycle. Much of the ocean's densest water vents to the atmosphere in the Southern Ocean, and changes in the rate at which this venting occurs may substantially affect the Carbon cycle. In comparisons of coarse-resolution models simulating ocean uptake of carbon dioxide (for example Sarmiento et al, 2000) large discrepancies in the magnitude and location of carbon fluxes are found in the Southern Ocean. Furthermore, changes in Southern Ocean ventilation associated with anthropogenic climate change could play a huge role in the changing the uptake of anthropogenic carbon dioxide by the oceans (altering it by up to 50%). However, all of these results have obtained with very coarse resolution models, and the robustness of these results is not known. In general, biogeochemical cycling in the presence of eddies is very poorly understood.
3. We are not ready to tackle a global eddy resolving calculation. We do not have the computational resources, and it is more difficult from a physical standpoint. A hemispheric simulation can be constrained closer to reality (e.g., by way of boundary conditions at the equator) so that climate drift is minimized, and pervasive problems with simulating the North Atlantic (e.g., the so-called overflow problem) and the Arctic Ocean (the need for a model to cross the pole, and the need for an ice model) are avoided. (See also the section on 'future work' below.)

2.2 The role of eddies in the general circulation

Why bother with trying to resolve eddies? We are some way from being able to resolve ocean eddies in coupled climate models, so is there a need for such a study, and if so why use two models?

- The role of eddies on the oceanic general circulation is not well known, and constitutes one of the most important current problems in oceanography. It is known, for example, that high resolution appears necessary to properly capture the departure of the Gulf Stream from the coast at Cape Hatteras, and it seems very likely that eddies are important in the transport properties of the Southern Ocean. The early simulations of the FRAM project at relatively modest resolution by today's standards [0.5° (longitude) $\times$ 0.25° (latitude)] also point to the importance of eddies in the ACC.
- The *eddy parameterization* problem — that is, the parametric representation of the effects of eddies in coarse resolution ocean models — is extremely important for climate modelling. The MESO experiments provide a good testbed for examining existing and proposed theories in a realistic setting, something that heretofore has barely been attempted. The importance of this topic motivates our decision to initially perform experiments with time-independent forcing fields, making a clean separation between eddies and the mean flow.
- We use two models because this is the best way to ensure the robustness of the results. We know of no better way to look at the effects of topography, of vertical resolution, of boundary-layer effects, and of the effects of adiabaticity and vertical diffusion. Furthermore, we learn much about the models themselves.

2.3 Eddy ocean models in climate models

Should we expect coupled climate models to resolve ocean eddies in the future, and what should we do about it?

- Simulations on the decadal-to-century timescale will not, in the near term, involve eddy resolving or even eddy permitting ocean models. (But these eddies must be parameterized, as discussed above.) On the shorter timescale, e.g. the inter-seasonal to five years scale, then the next generation of computers may yield sufficient power to perform simulations with eddy permitting ocean models. It is important to gain experience with such models now, else we will be in a game of catch-up in the future. Other labs (e.g., Los Alamos, NCAR, Southampton, the Hadley Centre) are performing high resolution simulations now. Typically, their simulations are at a still higher resolution than we envision, but are run for a shorter time and they (thus far) have integrated their models just once, rather than performing a variety of experiments.
- Eddy permitting models, perhaps with a resolution of $1/2^\circ$, may demand a different approach to the parameterization problem than traditional non-eddy resolving models with a grid size of, say, 2° . The suite of integrations at various resolutions within MESO is well suited to exploring this.

3 Future Directions

There are three general directions we wish to pursue.

1. Exploring the effects of eddy parameterizations in medium resolution models, in particular at the resolution (of order 1°) likely to be used in the next coupled models used for both long (decadal and longer) and short term climate predictions. Appropriate parameterizations are likely to be a key feature in realizing the full potential of simulations at this resolution. The MESO experiments

are a good testbed for such parameterizations, and we have a number of new ideas about parameterizing eddies (e.g., Smith and Vallis, 2001) that should be explored. These experiments are not especially demanding of computational time, but are demanding of human time and creativity.

2. Higher resolution experiments with a similar configuration. Even at $1/4^\circ$ there is no guarantee that the eddy statistics have converged. Simulations at $1/6^\circ$ and higher will be revealing here. We can expect these to take 3-4 times longer than at $1/4^\circ$. Offset against this is a reduced need to perform multiple experiments and to use both MOM and HIM. As well as its scientific value, this configuration will be of use in pushing the boundaries of our computer system, since it could easily take advantage of hundreds of processors and multiple nodes.
3. More realistic global simulations. An important next step will be to perform a suite of experiments covering the globe, at resolutions from 1° to $1/4^\circ$ and higher. This requires, inter alia:
 - (a) A bipolar grid or similar solution to the pole problem.
 - (b) An ice model, which in turn demands a seasonal cycle.
 - (c) An improved bottom-boundary-layer parameterization, which appears necessary to get a better simulation of overflow problems in the North Atlantic.
 - (d) An improved upper boundary condition for calculating ocean-atmosphere fluxes. Thus far, we have used restoring conditions. Use of an anomaly-energy-balance model would improve the realism of this, without the complexity of a full atmosphere.

We do not see any of the above as significant (i.e., progress-halting) problems: the ice-model exists; progress is being made with the bipolar grid and the bottom-boundary layer parameterization; the EBM exists, although it is currently not in FMS form. Such a model then becomes useful tool for both climate variability studies and oceanographic problems — water mass formation, the oceanic response to global warming, nutrient dynamics, eddy effects on the general circulation, etc.

Of the above three items, item 2 is the easiest to implement, and we envision beginning this as soon as our suite at $1/4^\circ$ is complete. Regarding item 1, we envision a continuous exploration of this general topic with both present and future integrations, as well as with more idealized models. With item 3 we start to make a serious connection with the coupled climate modelling efforts going on in the lab. Taken together, these will be the main oceanographic (i.e., not involving fully coupled models, or data assimilation) efforts that will require significant computational resources in the near future.

4 Computational Resources

What computational resources are needed for the project, both now and in the future?

Our current suite of integrations is at $1/4^\circ$ horizontal resolution, and entails about 80 years of simulation with both MOM and HIM. We estimate that this will take about 4-6 node-months to complete on the Origin. A number of experiments can be performed in parallel, if the resources are available. At higher resolution (e.g., $1/6^\circ$, same vertical resolution) the computer time per model year approximately triples, but as stated above we would not envision performing as complete a suite of experiments at this resolution. On the other hand, the potential difficulties with

global eddy permitting and eddy resolving calculations, taken together with the rich payoffs that are expected to accrue, suggest that many experiments at medium-to-high (e.g., from 1° to $1/4^\circ$ resolution) will be needed, although we prefer to wait until our MESO suite is more fully analysed, and until the structure of the ocean component of the trunk coupled climate model is better known, before proposing specific experiments.

5 Summary

Simply taken in isolation, the MESO calculations would be a highly visible benchmark oceanographic calculation, in the same spirit as FRAM. We also regard them as precursors for the climate models of tomorrow. That is to say, we believe the project is important both for its oceanographic value *and* because of the longer-term payoff in constructing better ocean models for the next generation of coupled climate models.

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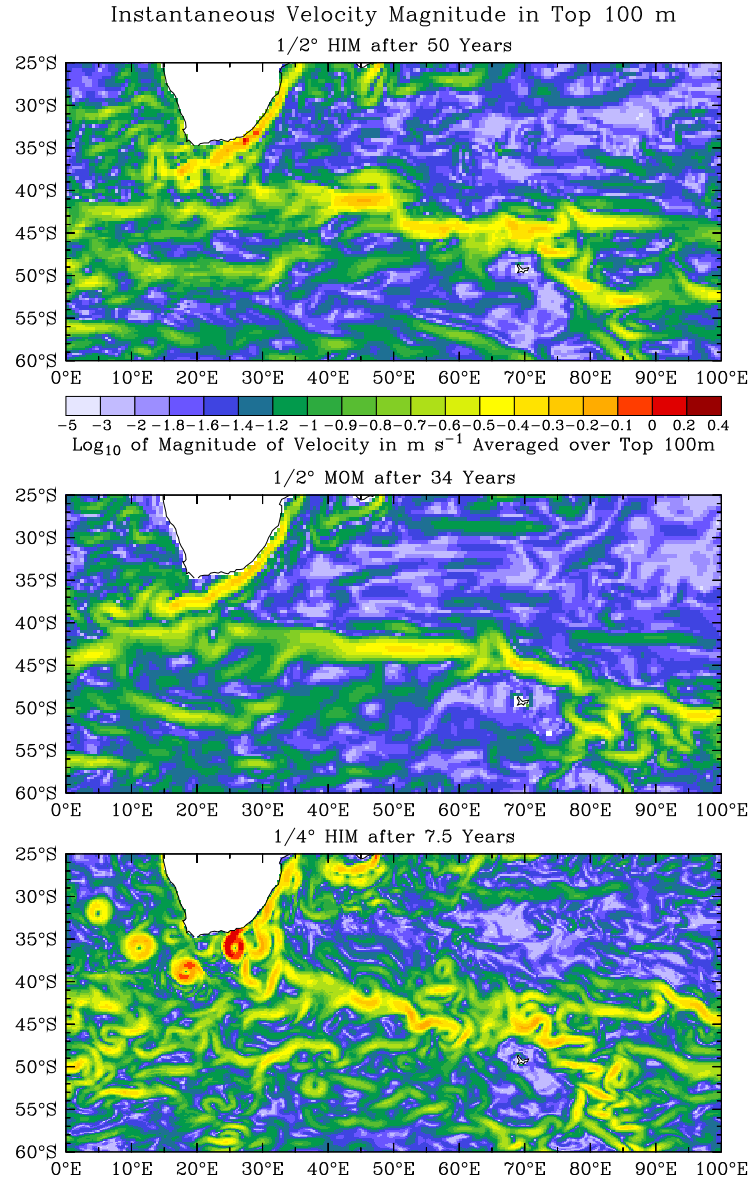


Figure 1: Circulation near the Agulhas at two different resolutions. Upper two panels: 1/2° , HIM and MOM. Lowest panel, 1/4° , HIM. Note that well-defined Agulhas rings appear only at the higher resolution.